# Integrated Circuit ECE326
# Lec. 12
# CMOS Processing Technology

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# Outlines

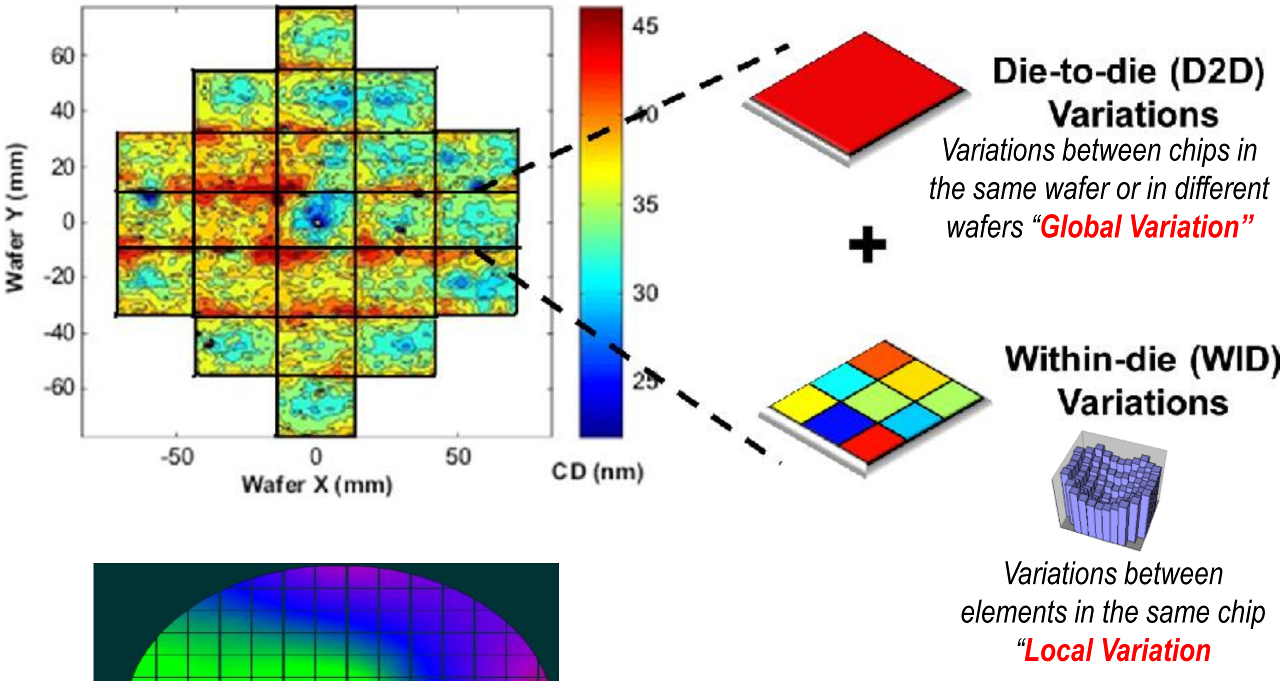
- ❑ Process Variations
- ❑ Manufacturing Issues
- ❑ IC Packaging

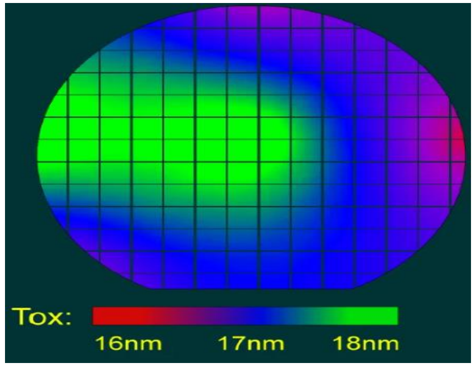
# Outlines

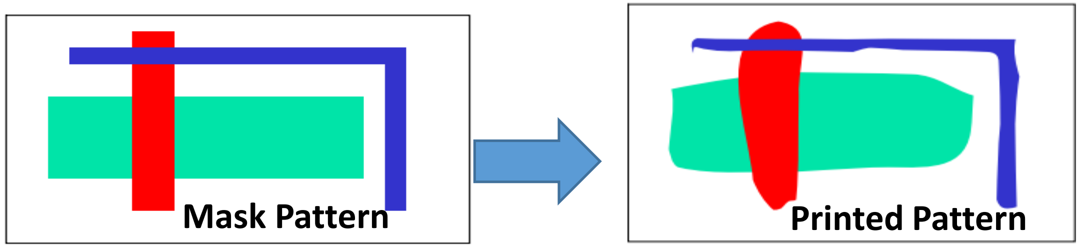
- Process Variations
- Manufacturing Issues
- IC Packaging

# Process Variations

- Variation can occur at different levels:
  - Fab to Fab variation
  - Wafer to Wafer variation
  - Die to Die Variation
  - Device to Device Variation
- Process Parameters
  - Impurity concentrations, oxide thickness, diffusion depth are varying during deposition and diffusion steps. Those parameter impacts threshold voltage ($V_{thg}$), Oxidation's thickness(tox)
- Device Dimensions
  - Lengths and widths of gates, metals, etc. due to photolithographic limitations.




*Those variations could be Random or systematic*

**Optical Proximity Correction (OPC) is required**

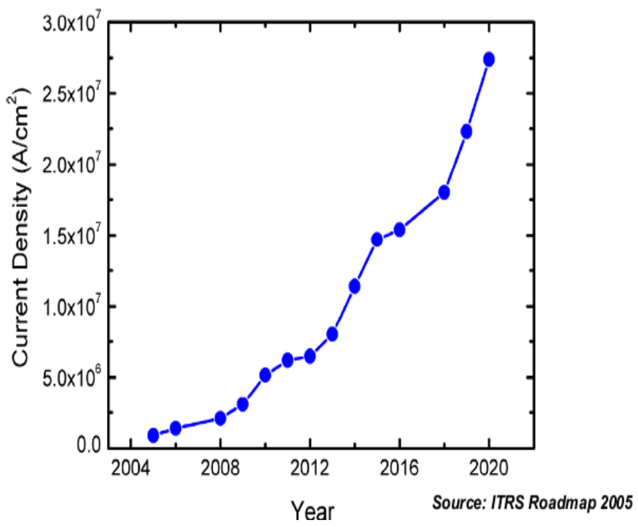
# Outlines

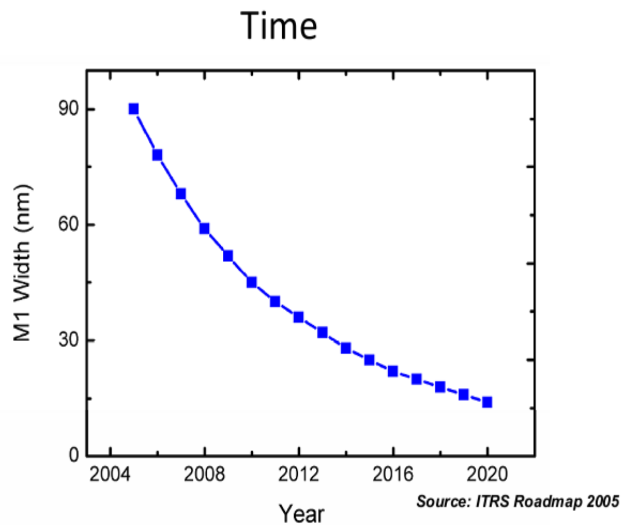
- Process Variations
- Manufacturing Issues
- IC Packaging

# Manufacturing Issues

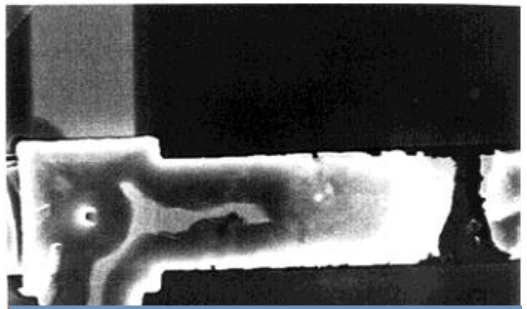
## ❑ Electro-migration (EM)

- ❑ EM is the gradual displacement of metal atoms in a semiconductor.
- ❑ The current density is sufficiently high to cause the **drift** of metal ions in the direction of the electron flow. That may cause **hillocks** (**shorts**) or **voids** (**opens**).

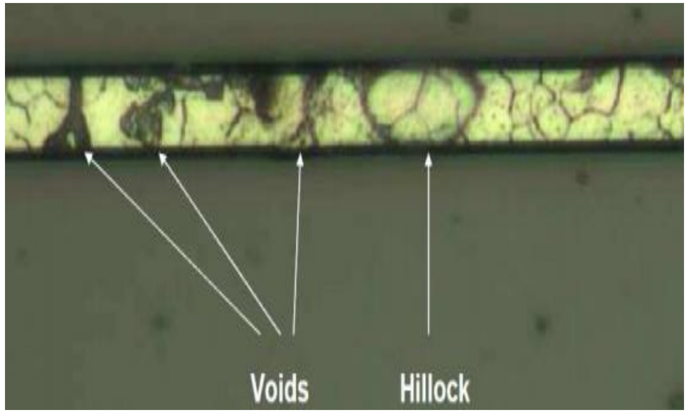



- ❑ **A hillock** causes a short to the adjacent or overhead metal runs. The reason is that the ions are piled up at a point where the incoming ion flux exceeds the outgoing flux.
- ❑ **A void** causes an open in the metal line because the flux of the outgoing ions exceeds the incoming flux



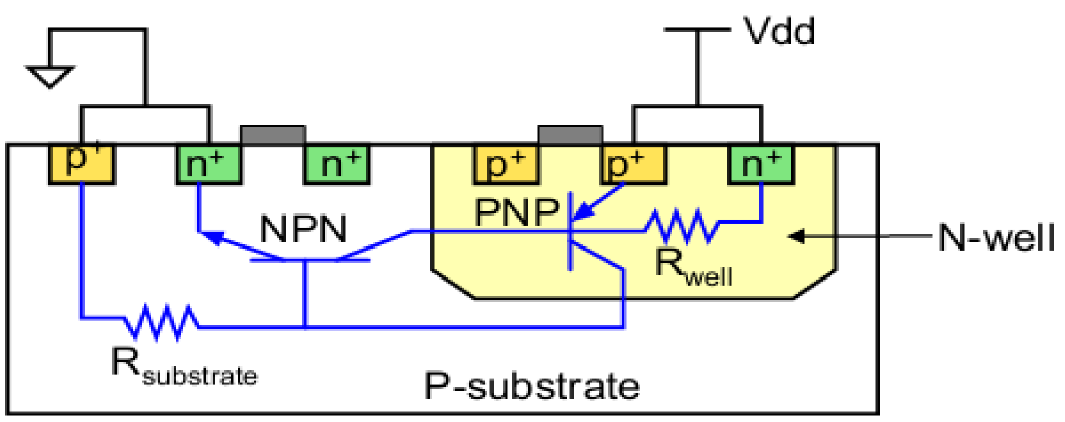
*A wire broken off due to electro migration*



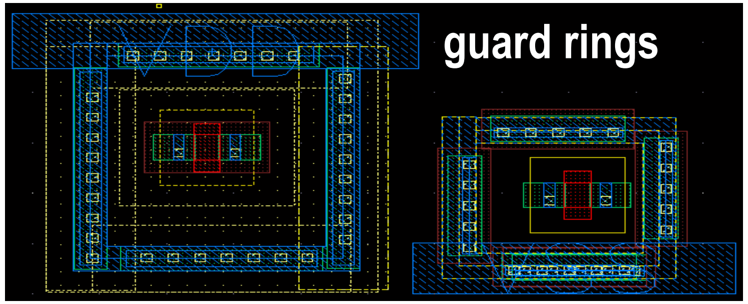
# Manufacturing Issues

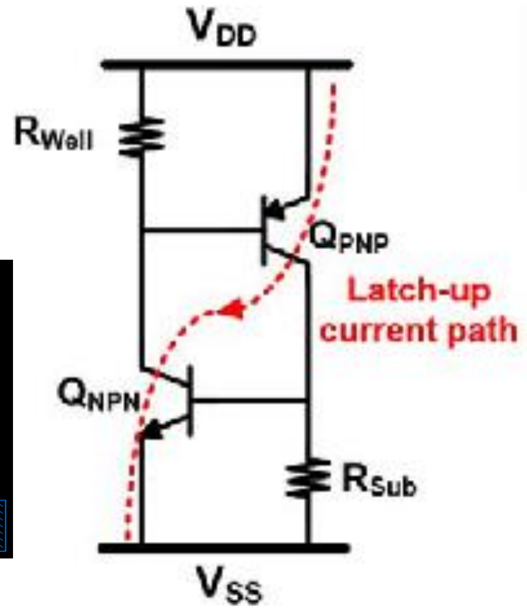
## ❑ Latch up

- ❑ Latch-up is defined as the generation of a low-impedance path in CMOS chips between **power supply rail** and **the ground rail** due to interaction of parasitic **pnp** and **npn** bipolar.




- ❑ **These BJTs form** a silicon- controlled rectifier (**SCR**) with positive feedback and virtually short circuit the power rail to, thus causing excessive current (**Damage** of the chip).



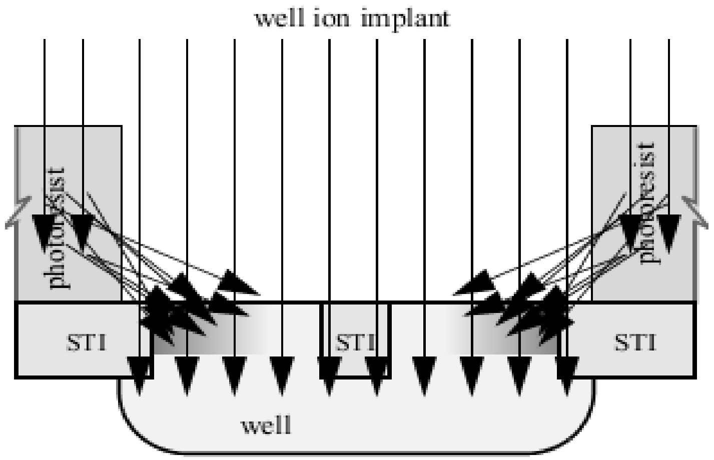



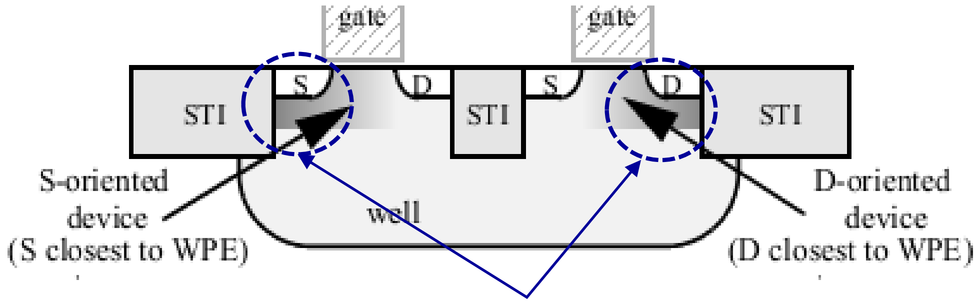

- ❑ To reduce the risk of latch-up, distribute well/substrate contacts across the chip using **guard rings**. Consequently, $\mathbf{Q_{npn}}$ and $\mathbf{Q_{pnp}}$ are “**Off**”

# Manufacturing Issues

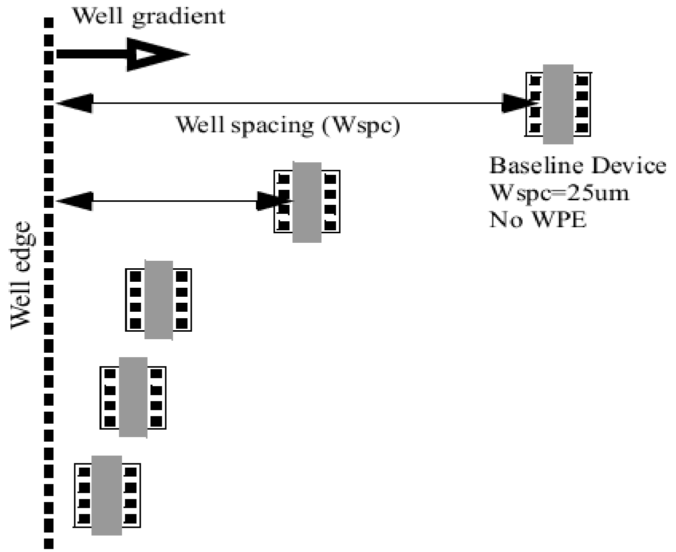
## ❑ Well proximity effect (WPE)

- ❑ During the implant process, atoms can **scatter** laterally from the edge of the photo-resist mask and become embedded in the silicon surface in the vicinity of the well edge.
- ❑ This lateral **non-uniformity** in well doping causes the MOSFET threshold voltages and other electrical characteristics to **vary** with the distance of the transistor to **the well-edge**.
- ❑ The **Vth** is found to increase by as much as 50mV as the device moves closer to the well edge which is consistent with previous observations. Many effort are done to detect the effect of **WPE** as shown.




**This region is not homogenous**

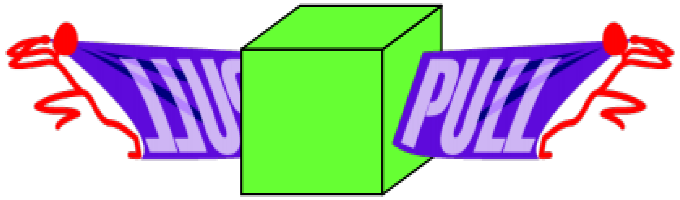


# Manufacturing Issues

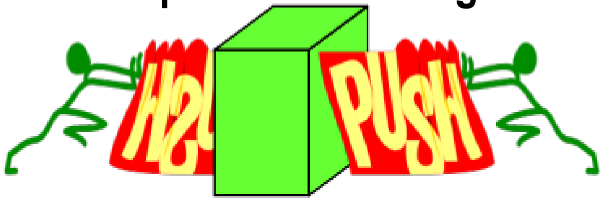
## ❑ Shallow Trench Isolation (STI)

- ❑ STI induced **stress** has been shown to have an impact on device performance introducing both $\mathbf{I_D}$ and $\mathbf{V_{th}}$ offsets.

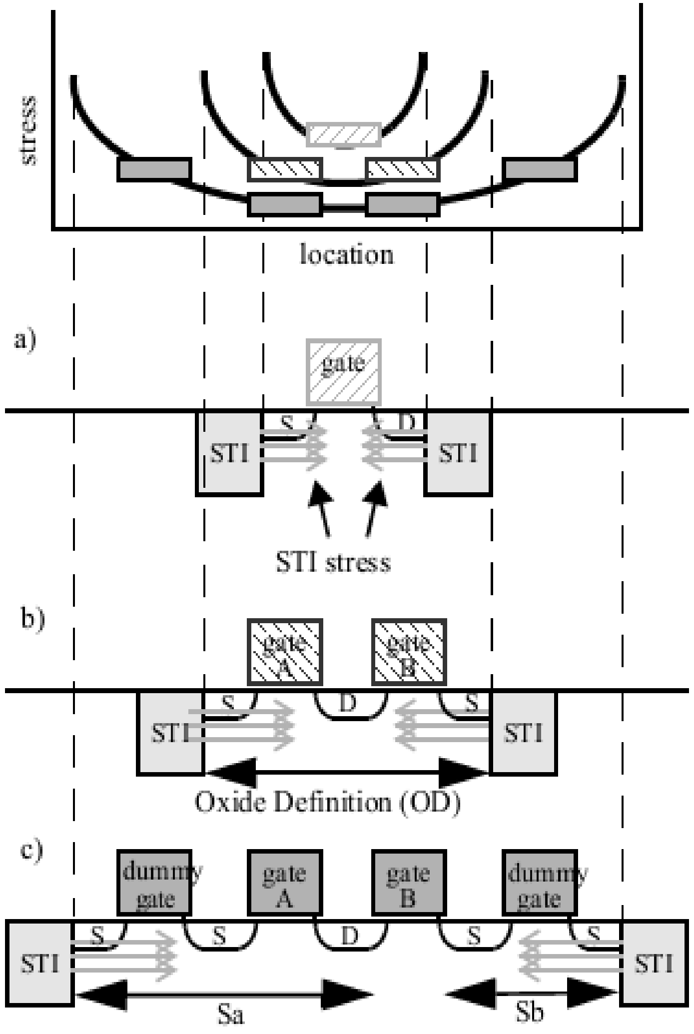
**Tension (positive stress)**
**stretch atoms apart**




**Compression (negative stress)**
**squeeze atoms together**




- ❑ It has been shown that the residual stress and corresponding shift in electrical performance can be qualitatively described by two geometric parameters, **Sa** and **Sb**.
- ❑ To mitigate the STI, **Dummy** gates would be used.




**The STI stress**

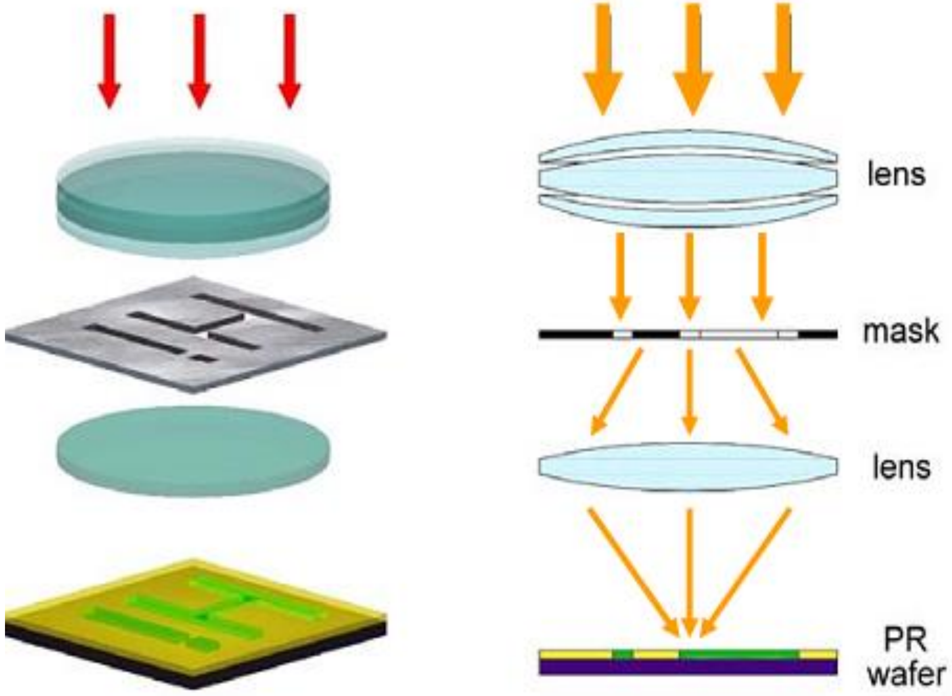
# Manufacturing Issues

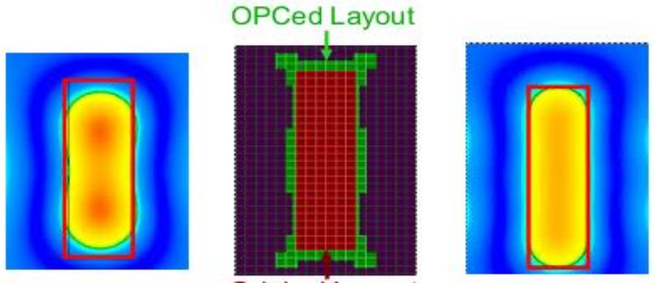
## ❑ Optical Proximity Correction (OPC)

- ❑ In microelectronics manufacturing, photolithography is the art of transferring pattern shapes printed on a mask to silicon wafers by the use of special imaging systems.

- ❑ Manufacturing sub-90 nm feature sizes requires intensive use of the **resolution-enhancement techniques (RET's)** to reduce the layout distortions.

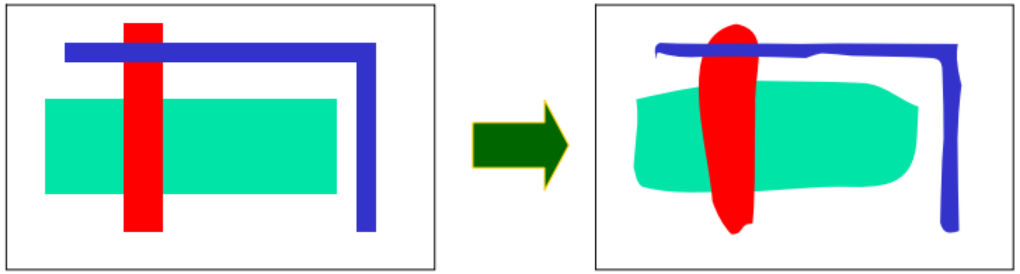
- ❑ **OPC** is the most popular RET in industry to improve the printed layout accuracy




Without OPC

With OPC

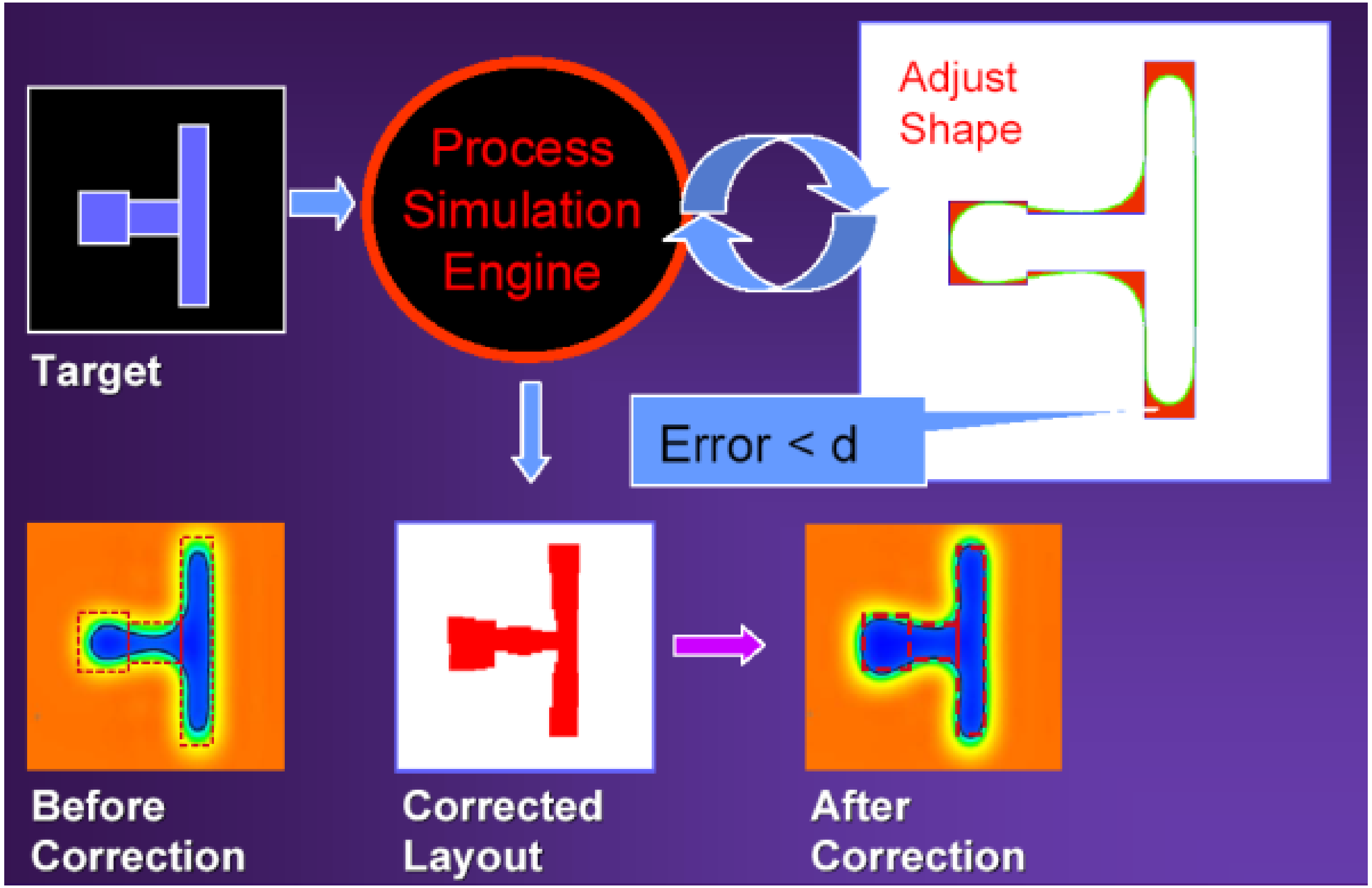


Mask patterns

Printed image

# Manufacturing Issues

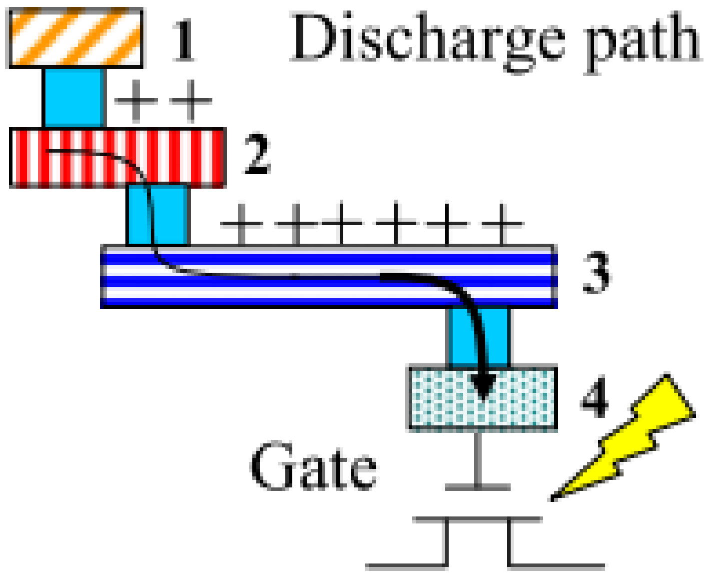
## ❑ Optical Proximity Correction (OPC)

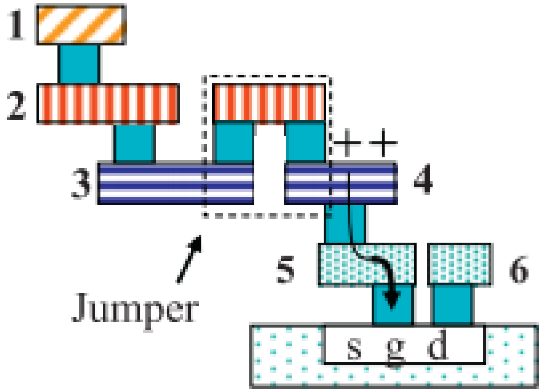


# Manufacturing Issues

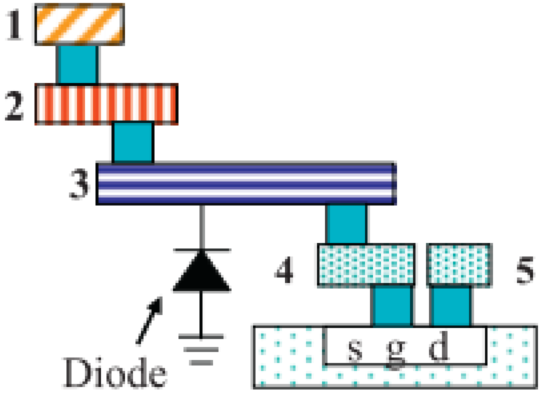
## ❑ Antenna Effect

- ❑ Charge is built up on interconnect layers during deposition.
- ❑ If enough charge is created, this can cause a high voltage to breakdown the thin gates.
- ❑ To eliminate the Antenna Effect, “Bridging/Jumper” or “Antenna Diodes” are used .




Reduce the antenna effect with jumper insertion



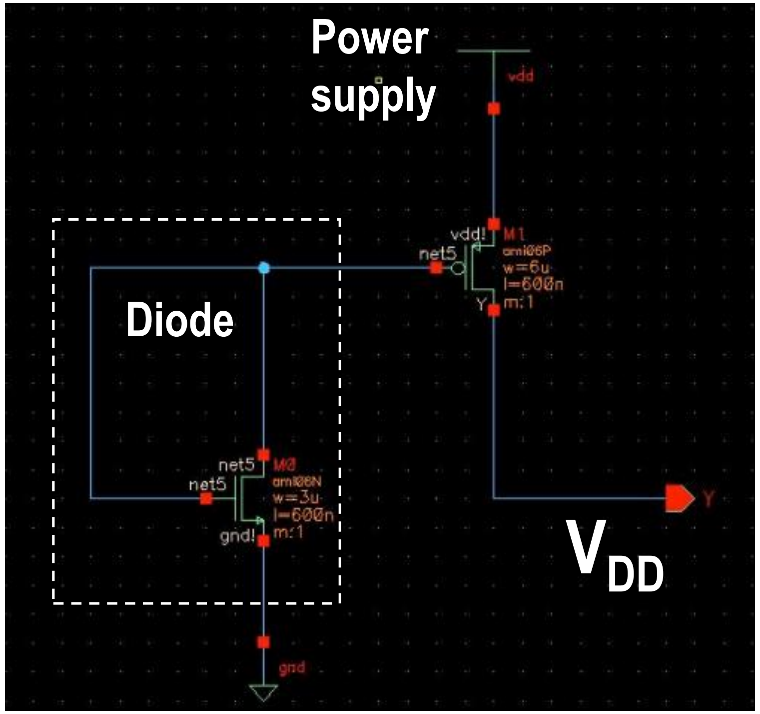

Reduce the antenna effect with Embedded diode

# Manufacturing Issues

## ❑ TIEHI and TIELO modules

- ❑ They provide high and low **Voltage Sources** to input of the MOSFET.
- ❑ They provide values for **gate** inputs that are tied to a **constant high or low voltage**.
- ❑ The reason is to protect the gate oxides from potential issues related to power supply spikes.

❑ The **TIEHI cell** uses a diode **connected NMOS** device that provides a low voltage to the gate of the PMOS transistor. Then, that PMOS provides a high voltage at its drain.



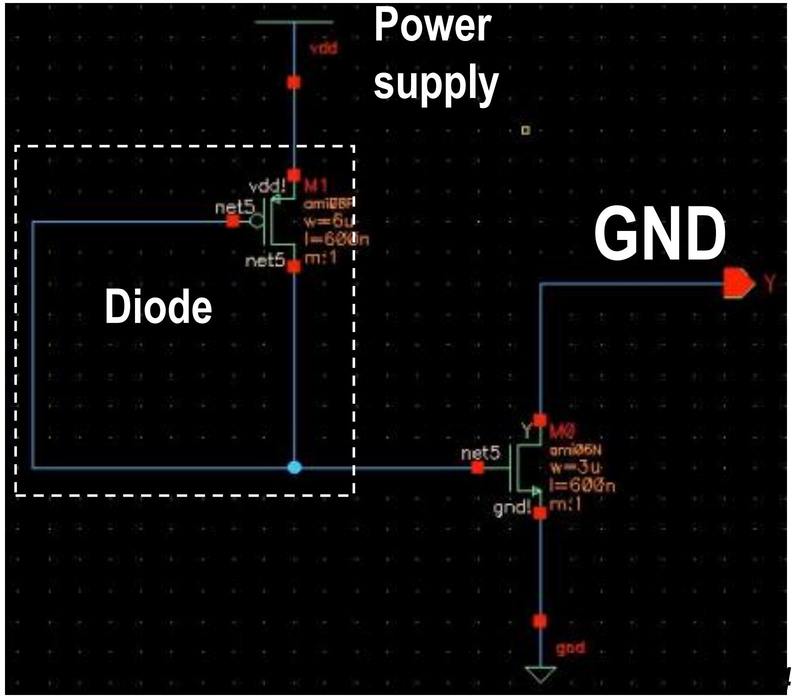

***A diode connected device is a transistor that has its drain and gate tied together so that it functions as a diode from drain to source***

# Manufacturing Issues

## ❑ TIEHI and TIELO modules

- ❑ They provide high and low **Voltage Sources** to input of the MOSFET.
- ❑ They provide values for **gate** inputs that are tied to a **constant high or low voltage**.
- ❑ The reason is to protect the gate oxides from potential issues related to power supply spikes.

❑ The **TIELO cell** uses a diode **connected PMOS** device that provides a High voltage to the gate of the NMOS transistor. Then, that NMOS provides a Low voltage at its drain



# Manufacturing Issues

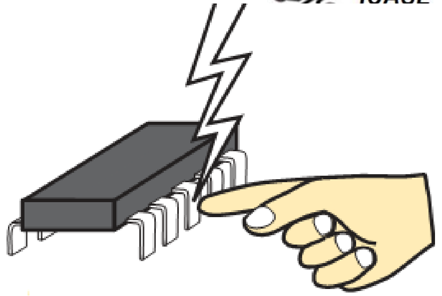
## ❑ Electrostatic Discharge (ESD)

- ❑ **ESD** is a transient discharge of static charge that **arises** from either human handling or a machine contact.
- ❑ The contact and separation of **feet** when **walking** across a floor creates a charge on the individual. (**15,000** volts of static electricity)
- ❑ The discharge to the doorknob is an example of an Electrostatic Discharge (ESD).
- ❑ The aggressive decrease in physical dimensions **results in** a significant decrease in gate-oxide thickness →Require less energy and lower voltage to avoid destroying MOS devices due to the current flowing through ICs during discharge.

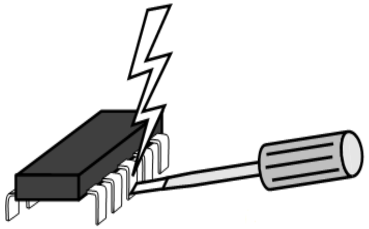
❑ Economically, ESD costs the electronics industry millions of dollars annually in damaged. Approximately **10: 60%** of device failure are ESD caused

# Manufacturing Issues

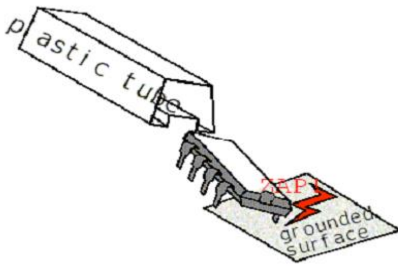
## ❑ Sources  and Model of ESD



❑ **Human body model** (HBM): This voltage is discharged when the person touches an object.



❑ **Machine Model (MM):** Equipment can accumulate static charge due to improper grounding. The charge is transmitted through ICs when it is picked up for placement in test sockets

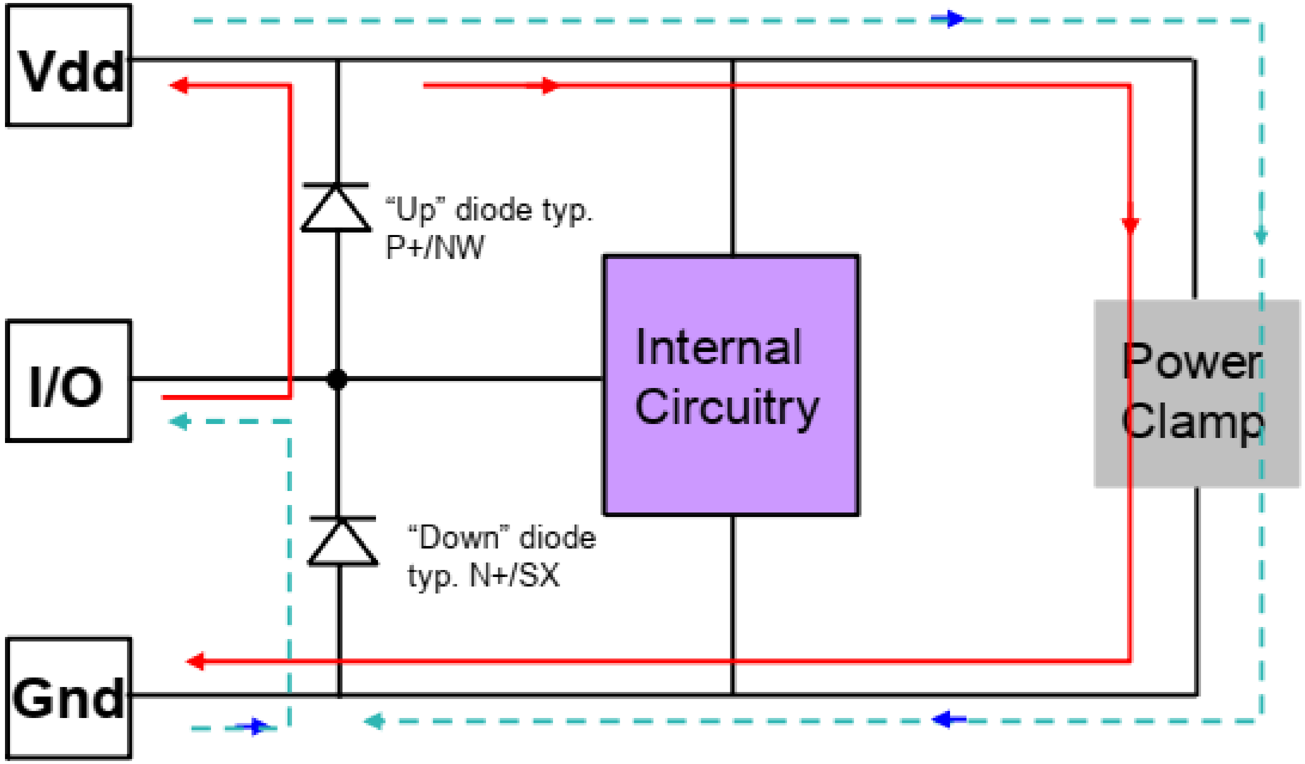



❑ **Charged device model (CDM):** ICs remain charged until they come into contact with a grounded surface (large metal plates /test sockets). Charge is discharged through the pins of ICs

| ESD Model | Industry Standard |
|---|---|
| Human Body Model (HBM) | 2000V |
| Machine Model (MM) | 200V |
| Charged Device Model (CDM) | 500V |

# Manufacturing Issues

## ❑ Typical Protection Circuit Topology – Double Diode Network




Discharge current paths in a protected design:

: Positive Pulse

: Negative Pulse

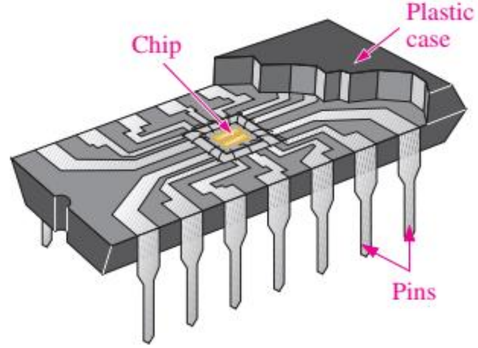
- ❑ Protection networks require a means of conducting discharge current from the supply to ground.
- ❑ The I/O voltage must remain between Vdd and ground (plus one diode voltage drop) or the diodes become forward-biased.
- ❑ A string of "up" diodes may be used to tolerate input voltages >Vdd.

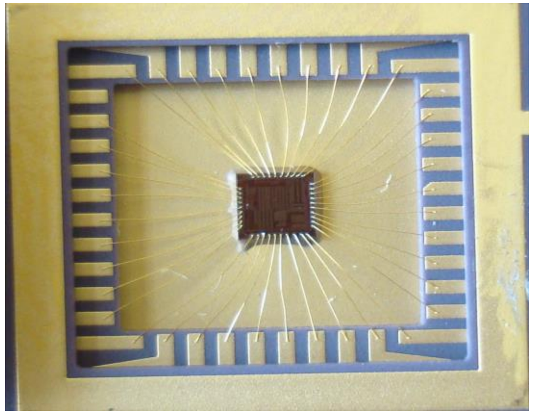
# Outlines

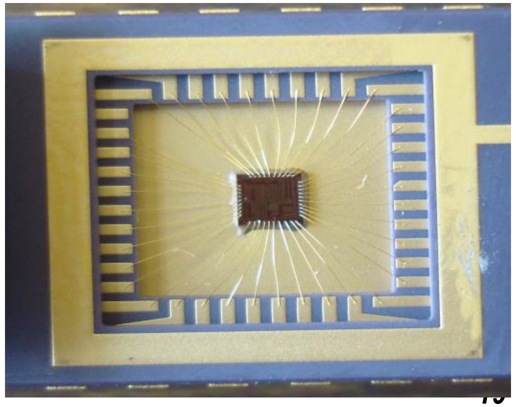
- ❑ Process Variations
- ❑ Manufacturing Issues
- ❑ IC Packaging

# IC Packaging

- **The package forms the interface between the silicon chip and the outside world and has a major impact on the performance, reliability, longevity and cost of the IC.**
- **Function of IC package are:-**
  - Protecting the die from the outside **environmental conditions**(**especially the humidity**).
  - Translating the on-chip interconnect density (um) to the off-chip interconnect density (mm).
  - Transferring the heat from the die to the outside world so it can be dissipated.
- **Requirements:**
  - Electrical → low parasitic capacitance, resistance and inductance.
  - Mechanical → reliable and robust, which requires a good matching of the coefficients for thermal expansion between the silicon and chip carrier and a strong connection from chip to package.
  - Thermal → efficient heat removal should be as high as possible
  - Economical → low cost is one of the most important properties



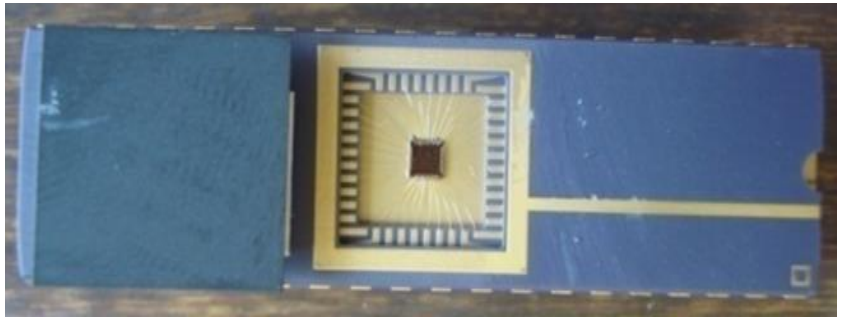


# IC Packaging

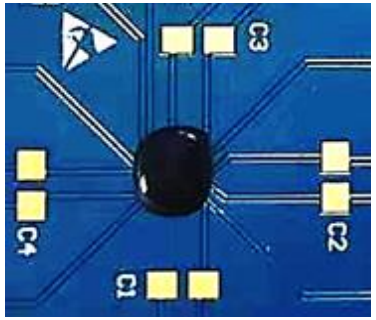
- **Packages can be classified in many different ways:**
  - By their main material
  - By the number of interconnection levels
  - By the means used to remove heat.

- The most common materials used for the package are ceramics and polymers (plastics).

- The polymers (plastics) is substantially **cheaper**, but has poor thermal properties.
- The ceramics has better thermal properties compared to plastic, but its high dielectric constant results in large interconnect parasitic.



**Ceramic**




**polymer**

# IC Packaging

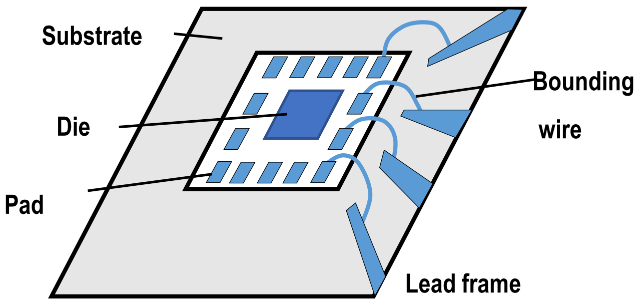
## ❑ Wire bonding

- ❑ The most widely used package interconnect is a wire bond.
- ❑ This is a thin gold wire that connects the pads on the IC to the package leads
- ❑ To accommodate a wire bond, we put pads around the perimeter of the IC
- ❑ The pads are relatively large (100um x 100um)

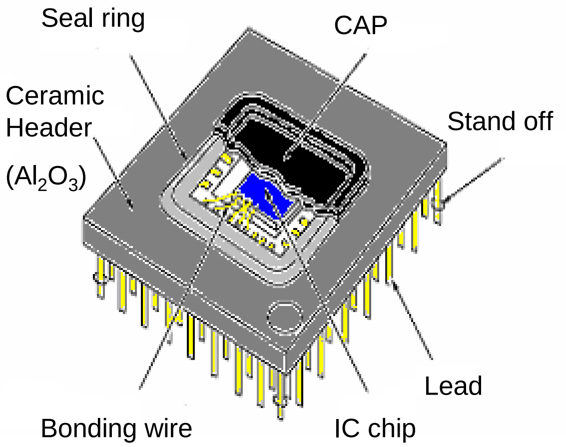
## Disadvantages:

- ❑ Lower throughput
- ❑ Bonding wires have high individual and mutual inductances.

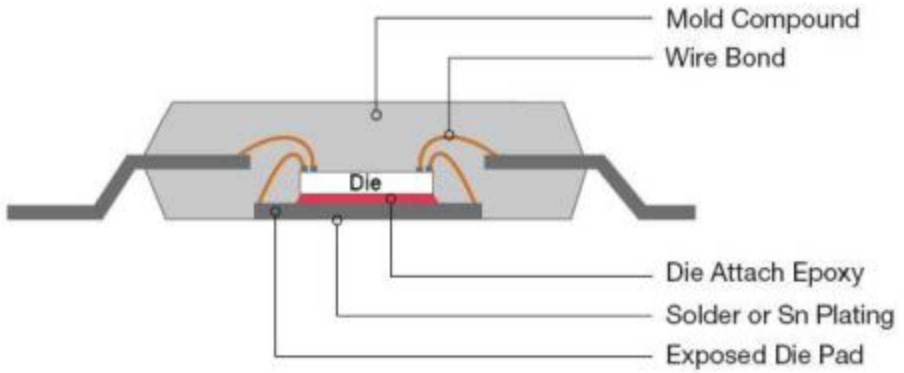
Wire Bonding




Ceramic Package Diagram




Wire-bond on a Lead Frame

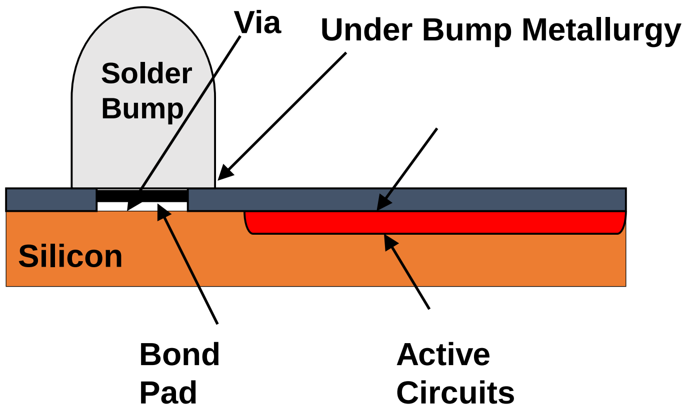


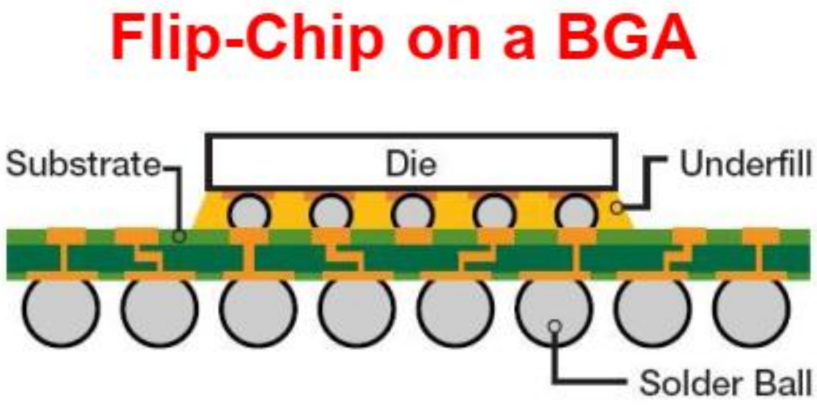
# IC Packaging

- **Flip Chip Bonding**
  - A higher performing and higher density interconnect
  - A sphere of solder is used to connect pads on the IC to the package

**The main advantages:-**

- Superior electrical performance
- The pads can be placed at any position on the chip
- Provide many more pads to be placed on the same die area



# Thanks!